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In-Field Testing of Digital Microfluidic Biochips using Multi-Agent Reinforcement Learning

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by

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CERTIFICATE

This is to certify that the work contained in this thesis entitled ***"In-Field Testing of Digital Microfluidic Biochips using Multi-Agent Reinforcement Learning"*** *is a bonafide work of* ***Ritik Kumar Koshta (Roll No. 234101044)****, carried out in the Department of Computer Science and Engineering, Indian Institute of Technology Guwahati under my supervision and that it has not been submitted elsewhere for a degree.*

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Guwahati.

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Abstract

Digital microfluidic biochips (DMBs) have emerged as a transformative technology in biomedical applications, enabling programmable droplet manipulation for tasks such as drug discovery, clinical diagnostics, and point-of-care testing. Among their many advantages, DMBs offer programmability, flexibility in fluidic operations, and versatile droplet mobility. However, their reliability can be compromised by factors such as manufacturing defects, electrode degradation, or dielectric breakdown, leading to incorrect fluidic behavior. Ensuring reliable operation is especially critical in safety-critical bioassays. In prior work, SAT solvers have been employed for in-field testing to determine optimal paths for test droplets. These droplets traverse the chip concurrently with ongoing bioassays to verify electrode functionality while maintaining assay integrity. Although SAT based methods are robust, their scalability is significantly limited by the computational complexity associated with larger grids and complex bioassays, resulting in prohibitive test completion times. To overcome these challenges, we propose a reinforcement learning (RL) based approach for optimizing test droplet routing in DMBs. By leveraging RL to learn efficient and reliable paths under predefined constraints, our method ensures rapid and effective in-field testing without disrupting ongoing assays. Experimental results demonstrate that the RL-based solution not only scales efficiently to larger grids but also achieves significant reductions in test completion time, thereby enhancing the reliability and scalability of DMB testing. This work represents a critical step forward in ensuring the dependability of DMBs in real-world biomedical applications.

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Chapter 1

Introduction

In recent years, advancements in microfluidic technologies have propelled digital microfluidic biochips (DMBs) to the forefront as the preferred platform for Lab-on-a-Chip (LoC) systems [1]. These platforms enable the execution of complex biochemical operations on a miniaturized, cost-effective chip by precisely manipulating fluids at nanoliter or picoliter volumes. DMBs perform a wide range of operations including droplet movement, mixing, and splitting via control signals that allow droplets to move one cell per time step [2]. This level of programmability significantly reduces human intervention and enhances automation in biological processes. Compared to traditional laboratory methods, LoC devices offer reduced reagent consumption, improved sensitivity, and the capability to support parallel DNA analysis, real-time biomolecular detection [3], and rapid clinical diagnostics [4, 5].

1.1 Microfluidic Technology

Lab-on-a-Chip (LoC) systems are broadly classified into two primary categories: Flow-based Microfluidic Biochips (FMBs) and Digital Microfluidic Biochips (DMBs). FMBs are typically multilayered devices—comprising flow and control layers—that manipulate fluids through a network of microchannels using pressure-actuated microvalves [6]. These platforms are generally tailored for specific applications and offer limited programmability. In contrast, DMBs have emerged as versatile, general-purpose liquid-handling platforms that manipulate discrete droplets at the picoliter scale via electrical actuation on a two-dimensional electrode array [7]. By altering electrode activation sequences, DMBs can execute a wide range of fluid-handling operations, including droplet dispensing, transport, mixing, and splitting. Numerous biochemical applications have been successfully implemented on DMB platforms, including nucleic acid analysis [8, 9], protein assays [10, 11], immunoreactions [12], and cellular assays [13]. Notably, the U.S. Food and Drug Administration (FDA) has approved the DMB based SEEKER platform (Baebies, NC) for clinical use in newborn screening for lysosomal storage disorders [14].

1.2 Digital Microfluidic Miochip DMB

A Digital Microfluidic Biochip (DMB) comprises a two-dimensional array of electrodes designed for executing biochemical reactions, along with peripheral reservoirs for reagent loading, as illustrated in Fig. 1.1(a). These devices utilize electrowetting-on-dielectric (EWOD) technology to perform various fluidic operations—including droplet dispensing from reservoirs, droplet transport, mixing, and splitting—through precise control of electrode activation sequences.

DMBs work on the basis of exact droplet movement that is regulated by outside influences. The majority of biochips including Electrowetting-on-Dielectric (EWOD) chips, manipulate the mobility of the droplets by applying electrical voltage to electrodes to change surface tension on the droplet which result in droplet motion. Similar control is obtained by other types of biochip such as Dielectrophoresis (DEP) or Surface Acoustic Wave (SAW) biochips, which employ electric fields or acoustic waves. Without the use of any external pumps or valves these chips make it possible to perform operations like mixing, dividing, dispensing, and conveying droplets.

Different types of Digital Microfluidic Biochips (DMBs) are differentiated by their droplet control mechanism:

- **EWOD (Electrowetting-on-Dielectric Biochips)**: They move droplets by regulating surface tension using voltage application.
- **DEP (Dielectrophoresis Biochips)**: They uses electric fields to move droplets that contain particles or cells.
- **SAW (Surface Acoustic Wave Biochips)**: They produce forces that results in droplets movement by using acoustic waves.
- **Magnetic DMBs**: They uses magnetic fields to control droplets movement that are for magnetically sensitive droplets.

An EWOD based Digital Microfluidic system typically consists of two parallel glass plates, as depicted in Fig. 1.1(b). The bottom plate incorporates a substrate with a patterned array of independently addressable electrodes, overlaid with dielectric and hydrophobic layers. The top plate features a continuous ground electrode coated with a hydrophobic film. Both the top and bottom plates are treated with hydrophobic coatings to reduce surface wettability, thereby facilitating controlled droplet manipulation.

A Digital Microfluidic Biochip (DMB) can be modeled as a grid graph $G = (V, E)$, where V represents the set of electrodes, and E denotes the set of edges that capture adjacency relationships between electrodes. A biochemical assay is typically represented as a directed acyclic graph (DAG), also referred to as a sequencing graph, where nodes correspond to fluidic operations and edges capture the precedence constraints among them. Fig. 1.2(a) illustrates an example assay involving the mixing of three fluids in a 1:2:5 ratio.

The sequencing graph is synthesized into an actuation sequence to realize the assay on a specific DMB. Fig. 1.2(b) depicts the symbolic actuation sequence for implementing the assay on a 7×7 DMB. The first line specifies the dimensions of the DMB, while the

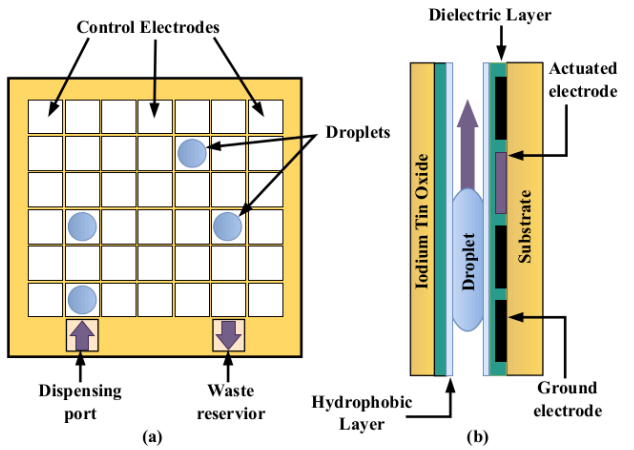

Fig. 1.1 (a) Schematic view and (b) Side-view of a general DMB. The droplet is moving to the right by EWOD.

next two lines define the locations of input, output, and waste reservoirs. The subsequent lines outline fluidic operations executed at each discrete time step t. For instance, the instruction `1 d([6,1]) d([6,7])` indicates that two droplets are dispensed at locations $(6, 1)$ and $(6, 7)$, respectively, at $t = 1$. Similarly, `2 m([6,1], [6,2]) m([6,7], [6,6])` represents the movement of droplets from $(6, 1)$ to $(6, 2)$ and from $(6, 7)$ to $(6, 6)$ at $t = 2$. The command `4 mix([6,2] [6,5], 8, 1x4) d([6,7])` specifies a mixing operation at locations $(6, 2)$ and $(6, 5)$ using a 1×4 mixer, which takes 8 time steps to complete. After mixing, two droplets remain at the same positions at $t = 12$, and a new droplet is dispensed at $(6, 7)$ at $t = 4$.

A snapshot of the DMB at time t is defined as the subset of vertices occupied by droplets at that time step. The *assay footprint* is the union of all such snapshots throughout the assay's execution, i.e., the set of all vertices that hosted droplets during the entire operation. Correspondingly, *footprint edges* are defined as the set of undirected edges induced by the footprint. As shown in Fig. 1.2(c), a droplet of reagent type B resides at vertex $(3,7)$ at $t = 7$ and moves to vertex $(3,6)$ at $t = 8$, thereby creating an undirected *footprint edge* $((3,7),(3,6))$, which represents the temporal transition of the same droplet between adjacent time steps.

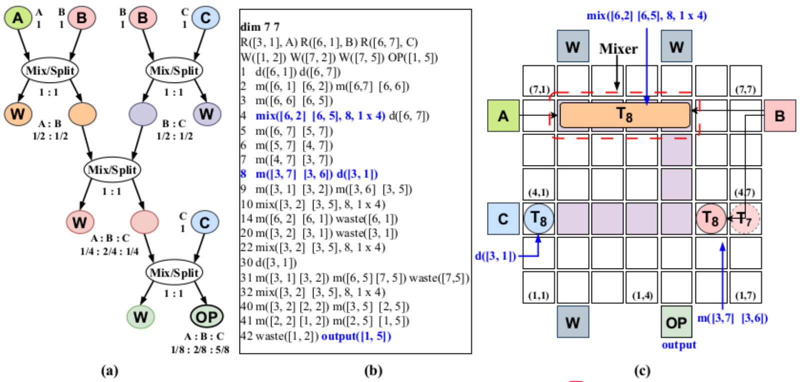

Fig. 1.2 Mapping of a Sequential-mixing assay in 1:2:5 on a 7 × 7 array: : (a) sequencing graph, (b) synthesized output, and (c) final mapping along with droplets position.

1.3 Defects and its Classification

The DMBs faces many challenges such as material fatigue, surface contamination, electrode failure and other which can possibily interrupt in the bioassays resulting in critical errors in experiment. As the size of biochips increases these issues occures more frequently. The challenge is to ensur that testing operates alongside with standard assays without interference enabling real-time defect detection while preserving bioassay integrity [15].

The droplet based (DMBs) are always prone to defects that can harm their functionality and reliability. These defects can be classified into three categories:

- **Permanent Defects**: These arise from irreversible physical damage or manufactur-

Table 1.1 Classification and examples of common defects in DMB's

Defect	Defect Type	Description	Examples/Manifestations
Permanent Defects	Catastrophic Faults	Defects that lead to complete and irreparable system malfunction.	Electrode short (e.g., between adjacent electrodes)
			Dielectric breakdown
			Electrode open fault (failure in electrode activation)
	Electrode-Specific Faults	Issues related to electrode fabrication or configuration that permanently impact functionality.	Misaligned electrodes
			Variations in metal deposition causing improper connections
Transient Defects	Fluidic Faults	Temporary issues caused by environmental or operational factors.	Particle contamination between plates causing high impedance
			Droplet stalling due to increased friction at the medium interface
	Parametric Faults	Environmental or fabrication deviations that degrade performance temporarily.	Increased droplet viscosity due to temperature variation
			Minor deviations in insulator thickness or electrode dimensions
Logical Defects	Edge-Dependent Faults	Issues in droplet routing or electrode interactions arising from logical or operational errors.	Errors due to electrode shorts along a specific flow direction
			Cross-contamination from residue left by previous droplets
	Routing Errors	Faults arising from improper or unintended droplet paths due to misconfigured routing logic.	Droplets taking unintended paths
			Collision of droplets during complex bioassay executions

ing flaws and they require hardware intervention.

- **Transient Defects**: These result from temporary environmental and operational conditions. They resolved once these conditions are addressed.

- **Logical Defects**: They originate from errors in the control logic or routing algorithms which leads to incorrect droplet behavior that can often be corrected via software adjustments.

Table 1.1 provides a detailed classification of these defect and categories along with respective examples [16]. This categorization shows the importance of understanding diverse defect types for enhancing biochip reliability.

1.4 Testing and its Types

Digital microfluidic biochips (DMBs) undergo with various types of testing to ensure their functionality and reliability, addressing potential defects that could impact performance during bioassay execution [17][7].

1.4.1 Offline vs. Online Testing

Based on when it is conducted the testing may be divided into two categories these are as following:

Offline Testing: Offline testing is conducted post-fabrication and prior to deployment it ensures that the biochip is free from physical and structural defects.

Extensive testing is performed to cover all electrodes and pathways of the biochip it makes sure it works as expected. This involves sending tiny drops of fluid through specific channels to check if the electrodes are functioning correctly. This approach is effective in identifying critical defects, such as shorts, dielectric breakdown, or open faults.

Purpose:

- To ensure that biochips are defect-free and reliable before being used for bioassays.
- Suitable for both disposable and reusable biochips.

Online Testing: Online testing is performed during real-time bioassay execution to detect and address defects dynamically without halting operations.

It actively monitors droplet operations to detect anomalies such as stalling or routing errors during real-time bioassay execution. Unlike offline testing, it does not require additional test droplets, as assay droplets are utilized for detecting anomalies. However, this method is limited to testing only the regions of the chip that are actively in use during the bioassay.

Purpose:

Table 1.2 Differences between structural and functional testing approaches

Aspect	Structural Testing	Functional Testing
Objective	Ensure physical integrity of the biochip by identifying defects in electrodes, connections, and pathways.	Validate the biochip's ability to perform fluidic operations like mixing, splitting, and dispensing.
Key Focus	Detect permanent and transient defects in structure.	Detect logical and transient defects during droplet manipulation.
Techniques	Graph-Based Models (Hamiltonian/Eulerian paths).	Real-Time Monitoring (tracks droplet movement).
	Predefined Droplet Paths (Routing test droplets).	Feedback Mechanisms (dynamic adjustments).
	Electrical Testing (voltage-driven tests).	Capacitive Sensing (detects droplet position and volume).
Defects Detected	Electrode shorts, dielectric breakdown, open faults, particle contamination, droplet stalling.	Routing errors, edge-dependent faults, stalling, irregular movement, volume errors.
Outcome	Ensures connectivity and operability of electrodes.	Ensures operational reliability during real-time bioassays.

- To ensure the correctness of droplet operations and mitigate errors during assay execution.
- Enables real-time fault detection and correction for improved reliability.

1.4.2 Structural and Functional Testing

Testing in digital microfluidic biochips (DMBs) can also be categorized based on what is being tested:

Structural Testing: Structural testing ensures the physical integrity of the biochip, verifying that all electrodes, connections, and pathways are free from defects.

Functional Testing: Functional testing validates the biochip's ability to perform fluidic operations like mixing, splitting, and dispensing.

The differences between structural and functional testing are summarized in the table 1.2.

1.5 Context and Purpose of the Study

Digital Microfluidic Biochips (DMBs) are susceptible to failures arising from both manufacturing defects and operational anomalies. Common manufacturing defects include non-uniform dielectric coatings, electrode shorts, misalignment of parallel plates, and broken connections to control pins. Operational failures may occur due to factors such as the application of excessive actuation voltage, which can result in dielectric breakdown and droplet-electrode shorts, or prolonged electrode activation, which may cause the electrode to become permanently stuck in the "on" state. Additionally, harsh environmental conditions—such as mechanical shocks leading to plate misalignment—or contamination from biological samples (e.g., protein adsorption and residue deposition) can impair droplet manipulation. Given that DMBs are frequently utilized in safety-critical biomedical applications, ensuring the reliability of fluidic operations is paramount.

To ensure robust performance, DMBs must undergo comprehensive post-manufacturing testing to detect and isolate defects prior to deployment. Several offline testing approaches have been developed, which involve routing test droplets across the electrode array to achieve high coverage of potential manufacturing defects while minimizing test application time. However, certain defects may remain latent and only manifest during in-field operation. In such cases, one approach to detect these errors involves routing test droplets in parallel with assay execution to traverse all adjacent electrode pairs (i.e., the assay footprint) visited by functional droplets. Designing an effective test-droplet routing schedule that avoids interference with functional droplets poses significant challenges. Bhattacharjee et al. [18] introduced a satisfiability (SAT)-based method to compute an optimal test plan—including routing paths and schedules—to achieve complete footprint coverage. However, the SAT-based approach is computationally intensive and thus practical only for small-scale assays on limited-sized DMB grids.

To overcome the limitations of SAT-based methods and enable scalable in-field testing, we propose a reinforcement learning (RL)-based framework that learns an effective routing plan for test droplet(s) to cover the assay footprint without interfering with the ongoing assay on the DMB. The RL agent (i.e., the test droplet) is capable of efficiently discovering in-field test strategies for large DMB grids in significantly less time.

The key contributions of this work are as follows:

- We introduce, for the first time, a tabular Q-learning-based reinforcement learning agent to derive a near-optimal in-field test routing plan.

- We conduct extensive simulations across a variety of assays and DMB grid sizes, demonstrating that the proposed Q-learning agent can rapidly generate near-optimal test plans for DMB grids as large as 50×50, where existing SAT-based methods fail to compute solutions within reasonable time limits.

Structural testing is applicable both in online settings (for real-time validation) and offline settings (for post-manufacturing defect detection). Functional testing, on the other hand, is primarily used online to ensure operational correctness during assay execution, though it can also serve as an offline method for validating specific assay functionalities.

This chapter provided a foundational background for the topics covered in this report. It begins with an overview of microfluidic technology, highlighting its significance and applications. We then dived into the principles of digital microfluidic biochips (DMB), exploring their structure and functionality. A classification of defects in DMBs is presented, followed by a discussion on various testing methodologies, including offline versus online testing and the distinction between structural and functional testing. The rest of the report is organized as follows: Chapter 2 presents a review of prior research, offering insights into existing approaches and the motivation behind this study. Chapter 3 covers essential reinforcement learning concepts, including fundamental elements, agent-environment interactions, Markov Decision Processes (MDPs), policies, and learning algorithms. Chapter 4 applies reinforcement learning to in-field testing, discussing problem formulation, modeling testing as an MDP, single-agent training, and multi-agent solutions. Chapter 5 presents the experimental setup, details results obtained through our proposed methods, discusses their implications and finally, summarizes the key findings and outlines potential future research directions.

Chapter 2

Review of Prior Works

The automation of biochemical processes in drug development, genome research, and clinical diagnostics has drawn interest in digital microfluidic biochips, or DMBs. However, flaws like electrode deterioration and dielectric breakdown raise serious concerns about their dependability. To solve these problems, a number of testing approaches have been put out over time.

2.1 Prior Work

The automation of biochemical processes in drug development, genome research, and clinical diagnostics has drawn interest in digital microfluidic biochips, or DMBs. However, flaws like electrode deterioration and dielectric breakdown raise serious concerns about their dependability. To solve these problems, a number of testing approaches have been put out over time.

Offline testing techniques were designed to detect physical defects in DMBs immediately after manufacturing. Su et al. [19, 20] and Mitra et al. [21] proposed methods for structural testing to ensure that all cells and boundaries in the microfluidic array function correctly. These techniques involved using Eulerization to generate efficient paths for test droplets to traverse all electrodes. Offline testing is effective in identifying manufacturing defects but does not address faults arising during runtime operations.

Mitra et al. [22] developed an online fault detection technique to guarantee dependability while the chip is operating. Without the need for extra test droplets, this technique dynamically repurposes assay droplets to identify problems during real-time execution. This method's coverage is constrained, nevertheless, because assay droplets can only follow the routes specified by the running assay and are unable to identify every physical flaw in the chip.

Concurrent testing was proposed to address the limitations of offline and online strategies by interleaving the testing process with bioassay operations. Su et al. [23] developed a Hamiltonian-path-based structural testing approach, where the objective was to traverse all cells in the microfluidic array while minimizing conflicts with functional droplets. However, this method focused on nodes (cells) and did not ensure complete edge traversal. Su et al. [20] reformulated the concurrent testing problem to focus on edge-based testing (Eulerian path), ensuring all cell boundaries between adjacent electrodes were traversed. While this approach improved defect coverage, it failed to address potential deadlock scenarios where test droplets could neither proceed nor stall.

Bhattacharjee et al. [18] suggested a SAT-based modeling technique to determine the best test schedules in order to overcome the difficulties associated with concurrent testing in digital microfluidic biochips (DMBs). This approach traverses all cell borders within the assay footprint, reformulating the in-field testing problem as a Boolean satisfiability problem (SAT) and guaranteeing robust, edge-based testing. In order to prevent conflicts between test and assay droplets, the SAT solver repeatedly calculates the minimal test application time. Due to the iterative nature of SAT solvers, this method involves a large computing burden even if it ensures scheduling free of deadlocks and robust fault detection. For real-time applications, this restriction creates a bottleneck, especially for complicated tests or big biochips.

Dinh et al. [24] have proposed a comprehensive testing methodology for DMBs that integrates practical constraints and optimizes test-application time but their approach has some limitations. The Integer Linear Programming (ILP) model, although capable of producing optimal schedules it is not scalable to larger biochip designs which restricts its applicability to modest layouts. The heuristic algorithm, though computationally efficient but it may not always achieve optimal results and serves primarily as an approximation to the ILP model's performance. Table 2.1 presents a comprehensive summary of previous research conducted in the field of testing DMBFs, including our proposed method.

Table 2.1 Characteristics of Various Test Approaches for DMBs

Test Approach	**Testing Type**	**Def. Detect.**	**No. of Test Droplet**	**Scalability**	**Approach Type**
Offline Testing [20][21]	Structural	Full	1–2 (heuristic)	till 15×15	PMF/CPP
Online error detection [22]	Functional	Partial	Nil	any size	Table-driven
In-field (concurrent) [23]	Structural (Hamiltonian)	Partial	1–2 (heuristic)	till 15×15	ILP
In-field (concurrent) [20]	Structural (Eulerian)	Full	1–2 (heuristic)	till 15×15	PMF (Heuristic)
In-field (concurrent) [18]	Structural (Eulerian)	Full	up to 4	till 15×15	SAT
Offline Testing [24]	Structural (Eulerian)	Full	any number	till 30×30	ILP/Heuristic
In-field (concurrent) **Proposed**	Structural (Eulerian)	Full	up to 4	till 50×50	Reinf. Learn.

Recent advancements have explored reinforcement learning techniques for droplet routing under fault conditions, especially in reconfigurable biochip platforms like MEDA. Elfar et al. [25][26] proposed deep reinforcement learning (DRL)-based approaches using PPO and CNN to dynamically route droplets while avoiding degraded micro-cells, enabling real-time performance and high scalability (up to 180×180). The earlier work focuses on adaptive rerouting based on local electrode health, while the later framework incorporates transfer learning for efficient deployment across varying chip layouts. Although these methods target MEDA biochips rather than conventional DMBs, they highlight the growing relevance of AI-based adaptive testing and routing strategies in microfluidic systems.

To enable high-throughput and fault-resilient droplet routing on MEDA biochips, Liang et al. [27] proposed a multi-agent reinforcement learning (MARL) framework where each assay droplet is autonomously controlled by a dedicated agent. The agents are trained using

PPO, ACER, and Double DQN algorithms to route droplets efficiently across degraded microelectrodes, relying solely on on-chip sensors without external cameras. The method supports parallel routing and is demonstrated on a real 60×80 fabricated MEDA chip, executing 49 concurrent droplet operations. Although targeted at MEDA platforms, this work showcases the power of MARL for dynamic, scalable droplet control in fault-prone environments

Kawakami et al. [28] proposed a deep reinforcement learning (DRL)-based droplet routing algorithm for digital microfluidic biochips (DMBs) that can dynamically adapt to both known and unknown faults. Unlike earlier approaches that rely on static sensor data, this method updates the routing policy during execution by detecting anomalies in droplet movement, allowing it to reroute around unexpected faults. Using a PPO-trained CNN agent, the algorithm achieves high success rates and near-optimal paths on DMBs up to 16×16 in size, without needing test droplets or camera-based feedback.

2.2 Motivation

The SAT-based method for in-field testing of digital microfluidic biochips (DMBs) provides reliable, edge-based defect detection and prevents deadlocks. However, its iterative SAT-solving process introduces considerable computational delays, making it less suitable for large biochips or time-sensitive applications. This paper introduces a reinforcement learning (RL) approach to significantly reduce test scheduling time, providing a faster, scalable, and efficient solution while preserving the robustness and precision of the SAT-based technique.

Chapter 3

Background

In this chapter, we explored the fundamentals of Reinforcement Learning (RL), a powerful approach in machine learning where agents learn to make decisions by interacting with an environment. We covered key concepts such as rewards, policy optimization, and value functions, highlighting how RL enables adaptive learning in dynamic scenarios. Additionally, we discussed practical applications, including robotics, game playing, and autonomous systems, illustrating RL's impact across various domains. This chapter serves as a foundation for understanding how intelligent systems refine their strategies through trial and error, ultimately achieving optimal decision-making.

3.1 Elements of Reinforcement Learning

Reinforcement Learning is a type of machine learning where an agent learns how to act in an environment to achieve a goal. It does so by interacting with the environment, taking actions, and receiving feedback in the form of rewards. The goal of the agent is to maximize the total reward over time by discovering the best actions to take in different situations.

- **Agent:** The decision-maker that learns by interacting with the environment to achieve a goal.

- **Environment:** Everything the agent interacts with, which responds to the agent's actions by providing new states and rewards.

Apart from the agent and the environment, a reinforcement learning system can be described through four key components: a policy, a reward signal, a value function, and a model of the environment.

- **Policy:** A policy in reinforcement learning defines the agent's approach to making decisions in a given state. It is essentially a mapping from states of the environment to actions, determining the behavior of the agent. The policy can be a simple function, such as a lookup table, or a more complex computation like a probabilistic model. For instance:
  - **Deterministic policies** always choose the same action for a given state.
  - **Stochastic policies** assign probabilities to actions, allowing for exploration.
- **Reward signal:** A reward signal is the feedback an agent receives from the environment after each action, guiding it to maximize total rewards over time. Positive rewards reinforce good actions, while negative rewards discourage bad ones, helping the agent adjust its decisions for better outcomes.
- **Value function:** A value function specifies the long-term desirability of a state by estimating the total reward an agent can expect from that state onward. Unlike rewards, which indicate immediate outcomes, value functions focus on future rewards by accounting for subsequent states and their potential.
- **Enviroment Model:** A model of the environment predicts the next state and reward for a given state-action pair, enabling planning by simulating future scenarios. Model-based methods use this for decision-making, while model-free methods rely on trial-and-error. Modern reinforcement learning often integrates both approaches.

3.2 Agent-Environment Interaction

The interaction between the agent and the environment unfolds in discrete time steps $t = 0, 1, 2, \ldots$:

- The agent observes the current state S_t.
- Based on a policy $\pi(a \mid s)$, the agent selects an action A_t.
- The environment transitions to a new state S_{t+1} and provides a reward R_{t+1}, determined by P_{sa} and R.

This process generates a trajectory:

$$S_0, A_0, R_1, S_1, A_1, R_2, \ldots$$

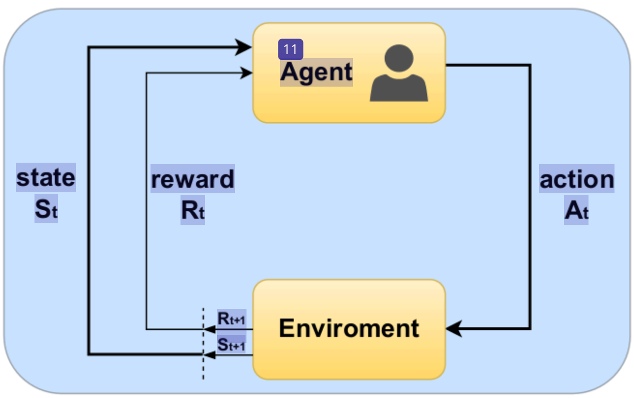

Fig. 3.1 Agent–Environment interaction in a Markov decision process.

Fig. 3 shows a flow diagram illustrating the interaction between states, actions, rewards, and transitions.

3.3 Markov Decision Process (MDP)

A Markov Decision Process (MDP) is a mathematical framework that models sequential decision-making problems where outcomes are influenced by both current actions and random factors. It provides optimal action mappings for each state under stochastic circumstances. An MDP is formally represented as a tuple: $(S, A, \{P_{sa}\}, \gamma, R)$, where:

- **State Space** (S): The state space S is the set of all possible states the environment can occupy. A state $s \in S$ contains all the information necessary to fully describe the environment at a given time step t. The state is used to determine both the available actions and the transitions to future states.
  - S is often finite or countably infinite.
  - The state captures only the relevant information needed for decision-making, following the Markov property.

  The **Markov property** states that the next state S_{t+1} and reward R_{t+1} depend only on the current state S_t and action A_t, and not on the sequence of prior states or actions.

- **Action Space** (A): The action space A is the set of all possible actions an agent can take. For each state $s \in S$, the agent can choose an action $a \in A(s)$, where $A(s) \subseteq A$ is the set of valid actions in state s.
  - Actions can be deterministic or stochastic.
  - The choice of an action affects both the immediate reward and the future state of the environment.

- **Transition Probabilities** (P_{sa}): The transition probabilities describe how the environment evolves based on the agent's actions. Specifically:

$$P(s', r \mid s, a) = \Pr(S_{t+1} = s', R_{t+1} = r \mid S_t = s, A_t = a)$$

This function defines the probability of transitioning to a new state s' and receiving reward r, given the current state s and action a.

- Transition probabilities satisfy:

$$\sum_{s',r} P(s', r \mid s, a) = 1 \quad \text{for all } s \in S \text{ and } a \in A(s).$$

  This property means that the total probability of transitioning to all possible next states (s') and receiving all possible rewards (r) from a given state (s) and action (a) must equal 1. This is because probabilities must account for all possible outcomes.

- Transition probabilities $P(s', r \mid s, a)$ encode the dynamics of the environment. This means they describe how the environment evolves based on the agent's actions and the likelihood of specific outcomes.

• **Discount factor (γ)**:The discount factor $\gamma \in [0, 1]$ determines the importance of future rewards relative to immediate rewards. It controls how far into the future the agent considers rewards when making decisions.

  - For $\gamma = 0$, the agent focuses only on immediate rewards.
  - For $\gamma = 1$, the agent considers all future rewards equally (for finite-horizon problems or episodic tasks).
  - Intermediate values of γ balance the trade-off between short-term and long-term rewards.

- **Reward Function (R)**: The reward function $R(s, a, s')$ assigns the expected scalar reward for transitioning from state s to state s' after taking action a. Formally, it is defined as:

$$R(s, a, s') = E[R_{t+1} \mid S_t = s, A_t = a, S_{t+1} = s'].$$

The equation expresses the expected reward (R_{t+1}) given that the agent is in state s, takes action a, and transitions to state s' at the next time step.

  - Rewards can be positive, negative, or zero, depending on the transition's outcome.
  - This function quantifies the immediate feedback the agent receives and is critical for learning and decision-making.

3.3.1 Objective

The agent's objective is to maximize the expected cumulative discounted reward, also known as the return:

$$G_t = R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \ldots$$

Or equivalently:

$$G_t = \sum_{k=0}^{\infty} \gamma^k R_{t+k+1}$$

This ensures the agent considers both immediate and future rewards, balancing short-term gains and long-term benefits.

3.4 Policies and Value Functions

- **Policy (π)**: A policy $\pi(a \mid s)$ defines the probability of taking action a in state s. The goal of reinforcement learning is to find an optimal policy π^* that maximizes the expected return.

- **State-Value Function ($v_\pi(s)$):** The expected return starting from state s and following policy π is defined as:

$$v_\pi(s) = E_\pi[G_t \mid S_t = s]$$

Expanded:

$$v_\pi(s) = E_\pi\left[\sum_{k=0}^{\infty} \gamma^k R_{t+k+1} \mid S_t = s\right]$$

- **Action-Value Function ($q_\pi(s, a)$):** The expected return starting from state s, taking action a, and then following policy π is defined as:

$$q_\pi(s, a) = E_\pi[G_t \mid S_t = s, A_t = a]$$

Reinforcement learning methods are categorized based on how the agent learns from interactions with the environment. On-policy methods, like SARSA, learn and improve the policy that the agent is currently following, balancing exploration and exploitation within the same policy. In contrast, off-policy methods, like Q-learning, allow the agent to learn about a separate target policy while exploring the environment using a different behavior policy. This distinction is critical in applications requiring flexible exploration strategies or learning optimal policies efficiently.

3.5 Optimal Policies and Value Functions

- **Optimal State-Value Function ($v^*(s)$):** The maximum value achievable from state s under any policy:

$$v^*(s) = \max_\pi v_\pi(s)$$

- **Optimal Action-Value Function ($q^*(s, a)$):** The maximum value achievable from state s, taking action a, and then following the optimal policy:

$$q^*(s,a) = \max_{\pi} q_{\pi}(s,a)$$

3.6 Q-Learning Algorithm

Q-Learning is a *model-free, off-policy reinforcement learning algorithm* used to learn the optimal action-value function, $Q^*(s,a)$, without requiring prior knowledge of the environment's dynamics. As an off-policy method, Q-Learning learns the optimal policy while following a potentially different behavior policy, such as ϵ-greedy exploration.

The goal of Q-Learning is to estimate $Q^*(s,a)$, which represents the maximum expected cumulative reward achievable from taking action a in state s and following the optimal policy thereafter. Using the Bellman optimality equation, $Q^*(s,a)$ is expressed as:

$$Q^*(s,a) = \max_{\pi} E\left[G_t \mid S_t = s, A_t = a\right],$$

where G_t is the cumulative reward.

The Q-Learning update rule is defined as:

$$Q(s,a) \leftarrow Q(s,a) + \alpha \left[R_t + \gamma \max_{a'} Q(S_{t+1}, a') - Q(s,a)\right],$$

where:

- $Q(s,a)$: Current estimate of the action-value function.
- α: Learning rate, controlling the step size of updates.
- R_t: Reward received after taking action a in state s.
- γ: Discount factor, weighing future rewards.
- $\max_{a'} Q(S_{t+1}, a')$: Estimate of the best future value from the next state S_{t+1}.

Q-Learning is guaranteed to converge to the optimal action-value function $Q^*(s, a)$ under the following conditions:

- All state-action pairs are visited infinitely often.
- The learning rate (α) decreases over time but remains positive.

As an off-policy algorithm, Q-Learning separates the behavior and target policies, allowing flexibility in exploration. It uses the Bellman optimality equation as a foundation and iteratively updates $Q(s, a)$ to converge to $Q^*(s, a)$, enabling the agent to learn optimal behavior in an unknown environment.

3.7 Applications of RL

Reinforcement learning is used in many domains, such as healthcare [29], robotics [30], autonomous control, and natural language processing (NLP) [31]. Its applicability also extends to scheduling, resource setup, autonomous systems, gaming, networking and communication, the Internet of Things, and computer vision.

Chapter 4

Reinforcement Learning based In-Field Testing

A potential technique for automating bioassays is the use of digital microfluidic biochips (DMBs), which manage tiny droplets on a range of electrodes. On such biochips, we propose a method to improve droplet routing using Multiagent Reinforcement Learning (MARL). The complex and dynamic environments seen in microfluidic systems are perfectly suited for MARL's ability to handle dynamic routing tasks in a flexible and adaptive manner.

The parallel droplet routing problem on a DMB may be defined using the MARL task. Each droplet is seen in this study as an independent agent that has to find its way to its target while dodging impediments and reducing the risk of contamination. By redefining the routing problem as a MARL job, we hope to decrease the training time to find optimal routing strategy so the problem of time consumption by SAT solver is solved.

In this section, we first formulate the in-field testing problem on DMBs. We then model the in-field testing problem as a Markov Decision Process (MDP), enabling the use of RL algorithms to find a test plan.

4.1 Problem Formulation

Input: A time-ordered sequence of droplet control commands corresponding to the execution of a bioassay on a specified digital microfluidic biochip (DMB).

Output: A detailed routing strategy for test droplet(s), comprising both the spatial path and temporal schedule, from designated source to sink reservoirs.

Goal: To determine a test droplet trajectory that:

- Traverses all footprint edges associated with the bioassay, where a footprint edge links two consecutive positions of the same functional droplet in adjacent time steps.
- Avoids spatial conflicts by ensuring that, at any given time step, the test droplet does not enter a 3×3 neighborhood centered around any active assay droplet.
- Minimizes the total time required to complete the test routing.

4.2 MDP-Based Modeling for In-Field Testing

The problem of in-field testing on a Digital Microfluidic Biochip (DMB) is modeled as a Markov Decision Process (MDP) defined by the tuple $\mathcal{M} = \{\mathcal{S}, \mathcal{A}, p, \mathcal{R}, \gamma\}$. Here, the agent (i.e., the test droplet) aims to learn an optimal routing policy that maximizes long-term rewards by covering as many assay footprint edges as possible, while avoiding collisions with functional droplets. The environment is deterministic, meaning every state-action pair leads to a unique next state, which is consistent with the predictable dynamics of droplet movement on a DMB.

4.2.1 State Space ($\mathcal{S}$)

Each state $s \in \mathcal{S}$ is represented as a triplet $s = (x, y, t)$, where (x, y) denotes the coordinates of the test droplet on the $M \times N$ DMB grid at time step t. The complete state space is defined as:

$$\mathcal{S} = \{(x, y, t) \mid x \in \{1, \ldots, M\},\ y \in \{1, \ldots, N\},\ t \in \{1, \ldots, T\}\},$$

where T is the total assay duration. Since the position of assay droplets at each time step is known a priori, incorporating time t into the state representation effectively encodes dynamic obstacles without explicitly expanding the state space to include droplet locations. This significantly reduces the state space size to $|\mathcal{S}| = M \cdot N \cdot T$.

4.2.2 Action Space ($\mathcal{A}$)

At any state, the agent can select one of the following actions: move left ($\leftarrow$), right ($\rightarrow$), up ($\uparrow$), down ($\downarrow$), or remain stationary ($\odot$). Formally:

$$\mathcal{A}(s) = \{\odot,\ \leftarrow,\ \rightarrow,\ \uparrow,\ \downarrow\}, \quad \forall s \in \mathcal{S}.$$

4.2.3 Transition Dynamics (p)

Transitions are deterministic. Given a state $s = (x, y, t)$ and an action a, the agent transitions to a unique next state $s' = (x', y', t + 1)$. For example, if the agent moves right:

$$p(s' \mid s, a) = \begin{cases} 1, & \text{if } s' = (x + 1, y, t + 1),\ a = \rightarrow, \\ 0, & \text{otherwise.} \end{cases}$$

4.2.4 Reward Function ($\mathcal{R}$)

The reward function is designed to encourage the agent to maximize footprint edge coverage while avoiding interference with the functional assay:

- A positive reward is given for traversing a previously uncovered footprint edge.
- A penalty is applied if the agent attempts to move into the 3×3 exclusion zone

centered on any assay droplet at that time step.

- A small negative reward is assigned for non-productive actions (i.e., actions that do not contribute to footprint edge coverage).
- Upon completion (at $t = T$), a final reward is given based on the overall percentage of footprint edges covered.

4.2.5 Discount Factor (γ)

A high discount factor $\gamma = 0.99$ is used to incentivize the agent to plan for long-term rewards, promoting exploration strategies that lead to higher cumulative edge coverage over time.

4.3 Single-Agent Solution for Testing

We employ tabular Q-learning [32] to train a test droplet agent for in-field testing on the DMB. The agent learns an action-value function $q(s, a)$ that estimates the expected cumulative reward for each state-action pair (s, a). The action-value function is maintained in a two-dimensional table $Q \in R^{|\mathcal{S}|\times|\mathcal{A}|}$, which is updated iteratively during training.

Algorithm 1 presents the Q-learning procedure tailored for this application. The input to the algorithm is the symbolic actuation sequence representing the assay. This sequence is parsed to extract the DMB dimensions $(M \times N)$, assay duration T, time-indexed snapshots of assay droplets, and the set of footprint edges $\mathcal{E}_f$ to be covered.

4.3.1 Initialization

Before training begins, the Q-table is initialized (typically with zeros), and hyperparameters such as the learning rate α, discount factor γ, exploration rate ϵ, and their respective update schedules are set.

4.3.2 Training Procedure

Each training episode proceeds as follows:

- The environment is reset by placing the test droplet at the initial position and clearing the set of visited footprint edges.
- At each time step t, the agent selects an action a using an ϵ-greedy policy [33]. With probability ϵ, a random action is chosen for exploration; otherwise, the best-known action according to the current Q-table is selected.
- The agent transitions deterministically to the next state s' based on the chosen action, receives an immediate reward r, and updates the Q-table using the Q-learning rule:

$$Q(s,a) \leftarrow Q(s,a) + \alpha \left[r + \gamma \max_{a'} Q(s',a') - Q(s,a) \right].$$

- The reward r is computed as:
  - $+15$ for covering a new footprint edge,
  - -20 for colliding with an assay droplet,
  - -1 for taking a non-productive action.
- If the agent completes the episode without any collision, a final reward (up to $+50$) is awarded based on the proportion of footprint edges covered during the episode.

An episode terminates under either of the following conditions:

1. The agent covers all footprint edges before time step T.
2. The agent collides with an assay droplet at any time $t \leq T$.

4.3.3 Exploration Strategy

The exploration rate ϵ is initially set to a high value to encourage exploration and is gradually decayed over episodes. A minimum threshold $\epsilon_{\min}$ is maintained to ensure continued exploration:

$$\epsilon \leftarrow \max(\epsilon \cdot \epsilon_{\text{decay}}, \epsilon_{\min}).$$

4.3.4 Learning Rate Schedule

The learning rate α is progressively increased during training using a controlled gain factor α_{gain}, up to a maximum value $\alpha_{\max}$:

$$\alpha \leftarrow \min(\alpha \cdot \alpha_{\text{gain}}, \alpha_{\max}).$$

This schedule allows the agent to make conservative updates early in training and gradually adopt more aggressive updates as it gains experience.

To demonstrate the RL-based in-field testing, an RL agent is trained for the mixing assay (Fig. 1.2(a)) running on the 7×7 DMB (Fig. 1.2(c)). The agent was trained with the following hyper-parameters: starting with learning rate $\alpha = 0.001$ we increase it by factor of $\alpha_{\text{gain}} = 1.00005$ till $\alpha_{\max} = 0.8$; discount factor $\gamma = 0.99$; and an initial exploration rate $\epsilon = 1.0$, which decays exponentially with a factor of $\epsilon_{\text{decay}} = 0.99995$ until reaching $\epsilon_{\min} = 0.01$. The training was carried out over 150 epochs, where each epoch runs for 1000 episodes.

Fig. 4.2 shows the average reward obtained by the agent increases and stabilizes as training progresses, indicating that the agent successfully learns a high-reward policy. Fig. 4.3 shows the average number of footprint edges covered by the agent also increases, showing high coverage with learning. To better visualize underlying trends, we plotted the maximum, minimum, and mean number of footprints covered at each epoch. These curves provide a clear representation of the performance range and average behavior throughout

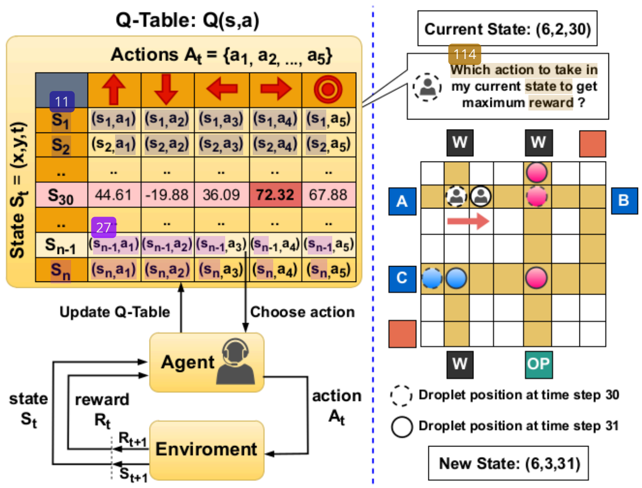

Fig. 4.1 Single agent Q-Learning architecture. The agent updates the Q-table during training (left) and uses the optimized Q-table to navigate the DMB during deployment (right).

the training process.

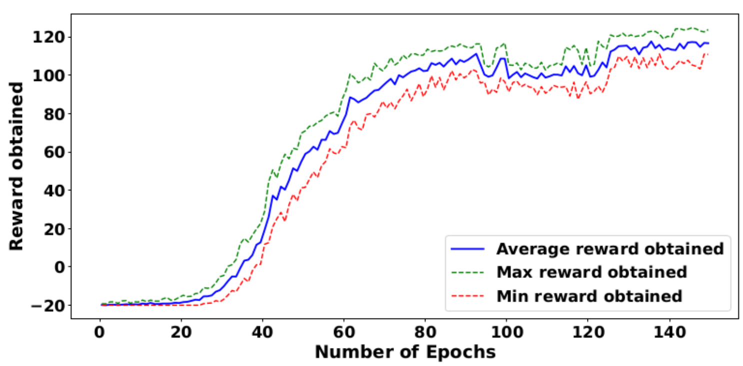

Fig. 4.2 Training performance of the agent on Sequential-mixing assay. The graph shows the reward agent obtained per epoche.

Algorithm 1: Q-Learning for In-Field Testing

Input: Actuation sequence
Output: In-field test plan
`// Training Setup`
Initialize $Q(s, a) = 0, \forall s \in \mathcal{S}, a \in \mathcal{A}$
Set hyper-parameters for learning $(\alpha, \alpha_{\text{gain}}, \alpha_{\text{max}})$, exploration $(\epsilon, \epsilon_{\text{min}}, \epsilon_{\text{decay}})$ and discount factor γ
`// Training continues for` n_{epochs}
for *(*$i = 1$; $i \leq n_{epochs}$; $i = i + 1$ *)* **do**
  for *(each episode in* i_{th} *epoch)* **do**
    resetEnv() `// reset test droplet position, covered footprint edges and time step`
    s = initState() `// initialize state`
    for *(each time step* $t = 1, 2, \cdots, T$*)* **do**
      a = selectAction() `// using` ϵ`-greedy policy`
      s' = takeAction(a) `// take agent to new state`
      flag = collisionCheck(s', t) `// agent-obstacle`
      r = getReward()
      $\text{TD}_{\text{error}} = \gamma \max_{a'} Q(s', a') - Q(s, a)$
      $Q(s, a) = Q(s, a) + \alpha \times \text{TD}_{\text{error}}$
      $s = s'$
      if *(all footprint edges are covered) or (flag)* **then**
        break `// terminate the episode`
      $\epsilon = \max\{\epsilon_{\text{min}}, \epsilon \times \epsilon_{\text{decay}}\}, \alpha = \min\{\alpha_{\text{max}}, \alpha \times \alpha_{\text{gain}}\}$
return $\pi(s) = \arg\max_a Q(s, a), \forall s \in \mathcal{S}, a \in \mathcal{A}$

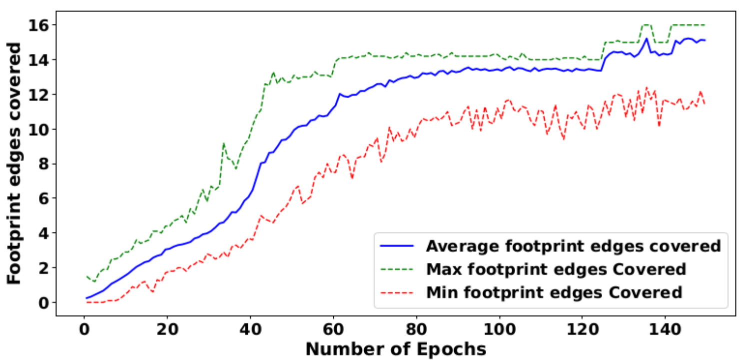

Fig. 4.3 Footprint edges covered per epoch by the agent during training in the Sequential-mixing assay.

Once training is complete, the learned Q-table is frozen, and the test droplet routing plan is represented as the final policy. At each time t, the agent selects actions according to the learned policy $\pi(s) = \arg\max_a Q(s, a)$ to route the test droplet without collision and cover footprint edges. Figs. 4.4(a)-(b) show the in-field test plan generated with the existing SAT-based approach and proposed Q-learning agent, respectively. The proposed method can cover all footprint edges by taking only two extra time steps compared to the optimal SAT-based method.

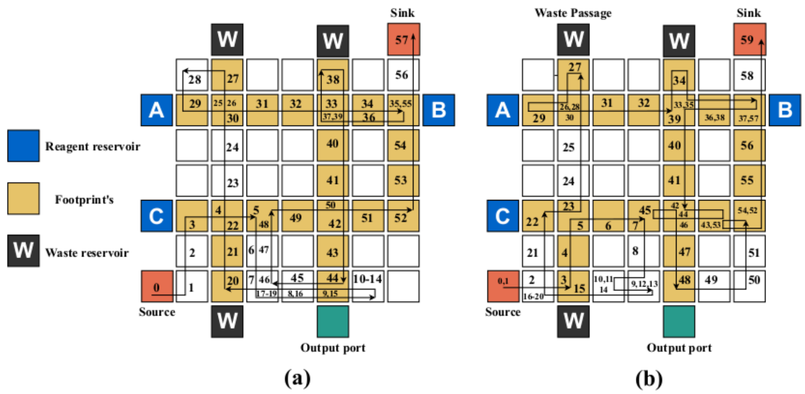

Fig. 4.4 Test paths obtained for the Sequential-mixing assay: (a) SAT-based method, (b) Proposed method.

4.4 Multi-Agent Solution for Testing

Instead of adopting conventional Multi-Agent Reinforcement Learning (MARL) paradigms such as centralized training, concurrent learning, or parameter sharing [27], we propose a decentralized variant of Independent Q-Learning (IQL) [34], specifically adapted for the footprint coverage problem on Digital Microfluidic Biochips (DMBs). Traditional MARL methods, especially those employing centralized training, encounter significant scalability bottlenecks due to the exponential growth of the joint state-action space with the number of agents.

To address this, our approach treats each test droplet as an independent reinforcement learning agent, maintaining its own Q-table and learning a local policy based on localized experience. Each agent operates primarily within a pre-assigned spatial quadrant of the DMB grid, promoting distributed and safe exploration. This custom MAQL strategy eliminates the computational burden of centralized coordination while preserving the benefits of parallel operation and decentralized learning.

4.4.1 Training Framework

Algorithm 2 extends the single-agent Q-learning algorithm to handle multiple agents operating simultaneously in a shared environment. Each agent A_i learns independently using IQL, maintaining a Q-table $Q_i(s, a)$ where $s = (x_i, y_i, t)$ denotes the agent's position and time, and $a \in \mathcal{A}$ is the action.

4.4.2 Quadrant-Based Partitioning

To promote even coverage and reduce inter-agent collisions, each agent A_i is assigned a unique quadrant $q_i \subseteq G$, where G is the overall DMB grid. An agent is rewarded only for covering footprint edges located within its designated quadrant. Although agents are permitted to move outside their assigned regions, no reward is provided for covering footprint edges beyond their designated quadrants. This spatial partitioning serves two primary objectives:

1. It reduces the probability of agent-agent collisions through implicit spatial separation.
2. It ensures balanced coverage across the entire assay area.

4.4.3 Reward Function and Safety Constraints

The reward function $\mathcal{R}$ is crafted to foster cooperative yet cautious behavior:

- $+15$ for covering a new footprint edge within the assigned quadrant.
- -20 for colliding with an assay droplet.

- -15 for colliding with another agent.

- -1 for a non-productive action.

Episodes terminate immediately if any agent collides with an assay droplet or another agent. If all agents survive until the end of the episode (at time T), each agent receives a final reward proportional to the collective footprint coverage:

$$\text{Final Reward} = 100 \times \frac{|\mathcal{E}_{\text{covered}}|}{|\mathcal{E}_{\text{total}}|},$$

where $\mathcal{E}_{\text{covered}}$ is the set of unique footprint edges covered by all agents.

4.4.4 Priority-Based Coordination

To prevent simultaneous collisions during execution, a lightweight priority-based action scheduling scheme is adopted. Agents are ordered based on the number of footprint edges in their respective quadrants, with agents responsible for larger quadrants receiving higher priority (e.g., $A_1 > A_2 > A_3 > A_4$).

At each time step:

- The highest-priority agent selects and executes its action first.

- Lower-priority agents observe the updated positions of higher-priority agents and choose actions accordingly.

- If a lower-priority agent's intended move leads to a collision, it remains stationary.

- In the case of an actual collision, the penalty is assigned to the higher-priority agent (initiator), encouraging cautious leadership.

This priority mechanism enables implicit coordination without centralized control or inter-agent communication, supporting scalable and safe multi-agent operation.

4.4.5 Post-Assay Greedy Completion Strategy

Upon completion of the assay (i.e., after time step T), both single-agent and multi-agent models adopt a greedy nearest-first approach to cover any remaining uncovered footprint edges:

1. Each agent selects the closest uncovered footprint edge (by Manhattan distance) within its quadrant.
2. This process repeats until all footprint edges are covered.
3. Finally, the agent navigates to its nearest sink location to conclude the task.

4.5 Optimistic Greedy Policy

In addition of ϵ-greedy behavior we have also used **Optimistic Greedy Policy** [35]. This policy encourages exploration by initializing all Q-values to high values, leading the agent to initially favor less-visited actions. As the agent interacts with the environment, the Q-values are refined, gradually reflecting the true expected rewards of actions and the agent shifts toward exploiting actions that yield the highest rewards. The initial optimism promotes broader exploration, which is particularly useful when the agent has minimal prior knowledge of the environment. This strategy ensures diverse behavior and thorough exploration before converging on the optimal policy. Additionally, once the learning process reaches a saturation point, a soft epsilon is maintained to allow occasional exploration, helping the agent avoid local optima and adapt to any dynamic changes in the environment.

Algorithm 2: MAQL for In-Field Testing

Input: Actuation sequence, number of test droplets k

Output: In-field test plan

// Training Setup

for *($j = 1$; $j \leq k$; $j = j + 1$)* **do**

  $Q_j(s, a) = 0, \forall s \in \mathcal{S}, a \in \mathcal{A}$

  Assign quadrant q_i

  Assign priority based on the number of footprint edges in q_i

Set hyper-parameters for learning $(\alpha, \alpha_{\text{gain}}, \alpha_{\max})$, exploration $(\epsilon, \epsilon_{\min}, \epsilon_{\text{decay}})$ and discount factor γ

// Training continues for n_{epochs}

for *($i = 1$; $i \leq n_{epochs}$; $i = i + 1$)* **do**

  for *(each episode in i_{th} epoch)* **do**

    resetEnv()

    // initialize state of each agent

    $s_j = \text{initState}()$, for $j = 1, 2, \cdots, k$

    flag = false

    for *(each time step $t = 1, 2, \cdots, T$)* **do**

      // for each agent

      for *($j = 1$; $j \leq k$; $j = j + 1$)* **do**

        $a_j = \text{selectAction}()$

        $s'_j = \text{takeAction}(a_j)$

        // check if agent j collides with higher priority agents $1, 2, \cdots, j - 1$

        if *(isCollisionBetweenAgents($j, 1, j - 1$))* **then**

          $a_j = stay$; $s' = \text{tickState}(s)$

          // 'tickState' keep agent position same while increment t

      for $j = 1$; $j \leq k$; $j = j + 1$ **do**

        // check if agent j collides with lower priority agents $j + 1, j + 2, \cdots, k$

        if *(isCollisionBetweenAgents($j, j + 1, k$))* **then**

          flag = true; $r_j -=$ penalty; // penalize the higher priority agent

        // 'isCollision' checks for collision between agents and functional droplets at t

        flag = flag *or* *isCollision*(s'_j, t);

        $r_j += \text{getReward}(s_j)$;

        $\text{TD}_{\text{error}} = \gamma \max_{a'} Q(s'_j, a') - Q(s_j, a_j)$

        $Q_j(s_j, a_j) \leftarrow Q(s_j, a_j) + \alpha \times \text{TD}_{\text{error}}$

        $s_j = s'_j$;

      if *((all footprint edges are covered) or (flag))* **then**

        break; // terminate the episode

    $\epsilon = \max\{\epsilon_{\min}, \epsilon \times \epsilon_{\text{decay}}\}, \alpha = \min\{\alpha_{\max}, \alpha \times \alpha_{\text{gain}}\}$

return $\pi_i(s) = \arg\max_a Q_i(s, a), \forall s \in \mathcal{S}, a \in \mathcal{A}, \forall i \in [1, k]$

Chapter 5

Experimental Setup and Results

This section outlines the system specifications, experimental setup, and results obtained from our reinforcement learning (RL)-based testing framework for digital microfluidic biochips. We also present a comparative analysis against the traditional SAT solver method to evaluate the efficiency and scalability of the RL approach.

5.1 Experimental Setup

The 46 experiments were conducted on a system running Microsoft Windows 11 Pro, equipped with an AMD Ryzen 5 3550U CPU, Radeon Vega Mobile Graphics, 8 GB RAM, and 476.9 GB dual SSD storage. This setup ensured smooth execution of both simulation and RL model training.

Table 5.1 summarizes the full set of experimental assays considered in this study. It includes both standard benchmark assays (originally analyzed using a SAT-based method) and newly constructed, more complex assays introduced in this work. The latter were designed by merging multiple baseline assays and executing them concurrently to simulate larger, more challenging testing conditions. These composite assays were specifically created to stress-test the scalability and robustness of our RL-based framework.

For all experiments reported in Table 5.1, the discount factor $\gamma = 0.99$ is kept constant.

The learning rate is set to $\alpha = 0.001$, with a gain factor $\alpha_{\text{gain}} \in [1.00003,\ 1.00005]$, where lower values are used for more complex assays. The maximum learning rate is bounded by $\alpha_{\max} \in [0.8,\ 0.9]$ with higher values applied to complex assays; The exploration rate is initialized at $\epsilon = 1.0$, with a decay factor $\epsilon_{\text{decay}} \in [0.99995,\ 0.999995]$; higher decay values are used for more complex assays. The minimum exploration rate is fixed at $\epsilon_{\min} = 0.1$. The number of training epochs ranges from 150 to 1000 depending on the assay complexity, with more epochs allocated to more complex assays.

Table 5.1 Summary of Assays used for comparing different approaches

Test ID	Assay Name	Dimension	T_{assay}
Assay 1	InVitro [18]	7 x 7	36
Assay 2	Sequential mixing of three reagents (Fig.1.2)	7 x 7	42
Assay 3	Multiple target mixture preparation: $A : B : C = 1 : 2 : 1$ and $A : C : D = 1 : 2 : 1$.	9 x 9	122
Assay 4	MinMix mixing [36] $R_1 : R_2 : R_3 : R_4 = 95 : 87 : 59 : 15$	11 x 11	192
Assay 5	Multiple target mixture preparation: $A : B : C : D : E : F = 1 : 1 : 1 : 1 : 2 : 10$, and $1 : 1 : 1 : 1 : 10 : 2$	13 x 13	155
Assay 6	Multiplexed InVitro [37]	16 x 16	160
Assay 7	Three instances of PCR mixture [38]	16 x 16	220
Assay 8	Plasmid DNA [36] mixing ratios = $(57 : 28 : 6 : 6 : 6 : 3 : 150)$	16 x 16	195
Assay 9	Multiplexed InVitro & Glucose [37, 20]	18 x 33	203
Assay 10	Mixing of six reagents: $A : B : C : E : G : H = 1 : 1 : 2 : 2 : 1 : 1$ and $A : B : D : F : G : H = 1 : 1 : 2 : 2 : 1 : 1$	30 x 30	184
Assay 11	Four instances of Multiplexed Glucose assays [20]	32 x 32	128
Assay 12	Combined execution of five complex assays—Assay 2, Assay 4, Assay 6, Assay 8, and Assay 10	50 x 50	533

5.2 Results

Table 5.2 presents a comprehensive comparison of all experimental assays evaluated using different testing approaches and agent configurations. For each test case, the number of agents, the total number of footprints, the test application time (TAT), and the running time are reported for three methods: SAT solver, ε-greedy reinforcement learning, and optimistic Q-learning. One point to note is that In the multi-agent setup with four agents, the SAT-based method utilizes a divide-and-conquer approach to simplify computation. On a larger grid, it reduces the number of boolean variables by dividing the space into four distinct regions, each assigned to a specific test droplet. The solution is then determined independently for each part using test droplets, making the problem more manageable. In contrast, reinforcement learning (RL) approaches—whether epsilon or optimistic greedy treat the problem as a unified whole rather than decomposing it. While agents are encouraged to remain within their designated quadrants for structured movement, they can also step outside if necessary to avoid collisions.

The TAT columns represent the total time taken to complete all test footprints using the specified strategy, while the running time columns indicate the computational effort (e.g., SAT solving or RL training time) required to generate the test path. Across various assays and grid sizes, RL-based approaches demonstrate competitive TAT—often near or slightly higher than that of SAT—while significantly reducing total computation time. Edge Coverage % refers to the percentage edges from the footprint that are successfully traversed by the algorithm within the allotted assay T_{assay} time. It serves as a key metric for evaluating the completeness and effectiveness of a solution in covering the designated spatial objectives. A higher Edge Coverage % indicates that the algorithm was able to explore a greater portion of the footprint within the time limit, reflecting better spatial coverage performance under time-constrained conditions. To ensure practicality, a cutoff time of approximately 3 hours was used for SAT-based experiments. If a solution was not obtained within this limit, the SAT solver was considered to have timed out, and we skipped further SAT evaluation for that assay and the TAT and Edge Coverage% are mentioned as NA for that particular assay on SAT solver.

In many scenarios, reinforcement learning incurs a modest increase in TAT due to its exploration-driven nature. However, it compensates by rapidly finding valid solutions, especially in larger or more complex grid environments. For example, SAT solvers may require 19,825.72 seconds to solve a 16×16 grid, whereas RL-based approaches can complete training and testing for a 16×16 grid in just 1695.78 seconds.

Among the two RL strategies, optimistic Q-learning consistently converges faster and achieves TAT values that are near similar to or slightly better than ε-greedy, while offering improved runtime performance. This makes it particularly suitable for real-time and scalable test generation in digital microfluidic biochips.

Table 5.2 Comparison of Experiments on Test Assays Across Methods

ID	T_{assay}	# Footprints	# Agents	Test Application Time			Running Time (seconds)			Edge Coverage %		
				SAT	ε-greedy	Opt. greedy	SAT	ε-greedy	Opt. greedy	SAT	ε-greedy	Opt. greedy
Assay 1	36	20	1	45	46	46	0.74	88.77	68.24	75.00	70.00	70.00
Assay 2	42	24	1	57	65	59	7.25	78.00	69.19	66.67	62.50	66.67
Assay 3	122	104	1	159	186	179	4831.05	1094.54	983.49	73.91	69.23	68.26
Assay 4	192	49	1	192	207	202	852.01	762.16	715.41	100	95.91	97.95
Assay 5	155	108	1	191	220	210	12698.28	1275.11	1201.97	88.04	83.33	85.18
Assay 6	160	319	4	160	207	182	19825.27	1842.81	1695.78	100.00	87.77	95.61
Assay 7	220	126	1	NA	256	252	Time Out	1396.39	1249.65	NA	92.06	92.85
Assay 8	195	115	1	NA	243	220	Time Out	1366.26	1197.09	NA	83.47	87.82
Assay 9	203	382	4	NA	277	239	Time Out	2058.34	1975.68	NA	84.03	97.12
Assay 10	184	261	4	NA	218	211	Time Out	2178.30	2094.44	NA	95.40	98.08
Assay 11	128	252	4	NA	171	151	Time Out	1673.85	1587.95	NA	84.92	90.47
Assay 12	533	768	4	NA	636	598	Time Out	8312.18	7893.27	NA	91.53	95.18

5.3 Conclusion

This study introduces a novel reinforcement learning (RL)-based framework for in-field testing of digital microfluidic biochips (DMBs), addressing the critical need for scalable and efficient fault detection in real-world biomedical applications. By dynamically computing optimal test paths that align with fluidic constraints, the proposed approach enables seamless concurrent testing alongside ongoing bioassays without interference.

The RL framework effectively leverages the fault detection model, identifying defects such as electrode malfunctions or droplet transfer failures with high precision. Experimental results underscore its superiority over traditional SAT-based methods, demonstrating significant reductions in test completion times and scalability to larger grid sizes and more complex bioassay operations.

This work establishes a robust, scalable, and efficient foundation for ensuring the reliability of DMBs, facilitating their adoption in advanced biomedical applications such as drug discovery, clinical diagnostics, and point-of-care testing. By overcoming the computational bottlenecks of existing methods, this approach represents a critical advancement in the dependability of DMBs, ensuring their long-term viability in safety-critical scenarios.

5.4 Future Scope

To enhance the scalability and adaptability of our model, our future research will focus on integrating advanced neural network-based methods. Specifically, we aim to explore Deep Q-Networks (DQN) and other reinforcement learning techniques to refine our approach. By leveraging these methodologies, we seek to improve the model's ability to dynamically adjust and optimize performance in complex environments, ensuring greater efficiency and flexibility in its applications.

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